Richard Feynman was an exceptional physicist and inspirational teacher who made the most esoteric of subjects—quantum theory—seem engaging, even entertaining. His subatomic ideas marked the beginning of a new era for physics and technology.

MILESTONES

WARTIME WORK
Appointed junior physicist in the Manhattan Project at Los Alamos, New Mexico, in 1943.

CAREER ASCENT
In 1950, becomes professor of theoretical physics at the California Institute of Technology (Caltech).

HIGH HONOR
Receives the Albert Einstein Award, a physics honor instituted while Einstein was still alive, in 1954.

POPULAR TEXTBOOK
His lectures at Caltech are published as the much-loved *Feynman Lectures on Physics* in 1964.

NOBEL WINNER
Awarded the Nobel Prize in Physics (jointly with two others) in 1965, for his QED work.

Born in New York to parents of Lithuanian Jewish descent, Richard Feynman did not learn to talk until he was 3 years old but had a talent for working with gadgets and showed a prodigious capacity for mathematics: in high school, he experimented with his own mathematical theories. Refused admission to Columbia University because it had already filled its entrance quota for Jews, he instead attended the Massachusetts Institute of Technology (MIT), studying first mathematics, then physics. In 1939, having gained his degree, he sat the graduate entrance exams for Princeton University and achieved a perfect score in physics—something no one had ever done before. He completed his PhD in quantum mechanics in 1942. During this time, he was asked to participate in the Manhattan Project: the Allied nuclear-weapons program. As he was concerned about the possibility of a Nazi victory in the war, he eventually agreed.

In 1945, Feynman was appointed professor of theoretical physics at Cornell University, where he started his most important work, on quantum electrodynamics (QED), an arcane field theory that had first emerged in the 1920s. QED described the interactions of electromagnetically charged particles in terms of an exchange of photons—packets of light, or "quanta"—of electromagnetic energy. However, there were significant mathematical flaws in the early

__Feynman conducted__ secret bomb research for the Manhattan Project at Los Alamos National Laboratory, New Mexico. He and many other scientists involved were accommodated in the site's vast trailer park.